

DA5000W Linear algebra and optimization

Assignment 4

December 2025

Topics:

1. Unconstrained Optimization
 - (a) First Order Condition
 - (b) Second Order Condition
2. Constrained Optimization
 - (a) KKT Conditions
 - (b) Equality and Inequality Constraints
3. Linear Program
4. Dual Problem

Question 1: Consider the function given below,

$$f(x) = x^4 - 4x^2, \quad x \in \mathbb{R}.$$

1. Find all stationary points using the first-order condition.
2. Classify each stationary point (local minimum, local maximum or saddle point) using the second-order condition.
3. Give the function value at each stationary point and identify the global minimizers.
4. Is the local maximum value a global maxima? Comment on it.

Question 2: Find the minimizer and minimum value of

$$F(x) = \frac{1}{2}(x_1^2 + x_2^2 + x_3^2)$$

subject to the single linear constraint

$$x_1 + 2x_2 + x_3 = 3.$$

Question 3: Let $x = (x_1, x_2)^\top \in \mathbb{R}^2$. Consider the problem

$$\text{minimize}_{x \in \mathbb{R}^2} \quad F(x) = \frac{1}{2}x_1^2 + 2x_2^2$$

$$\text{subject to} \quad A^\top x = b, \quad \text{where } A = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

- (a) Write the Lagrangian $\mathcal{L}(x, \lambda)$ for this problem.
- (b) Write the KKT stationarity equations.
- (c) Solve those equations to find x^* and the multiplier λ^* in terms of b.

Question 4: Using the methods given below, minimize the function

$$f(x, y) = (x - 1)^2 + (y + 2)^2$$

use $\alpha = 0.25$ and start from the initial point $(x_0, y_0) = (0, 0)$.

1. Compute the first three iterations of Gradient Descent.
2. Mention the optimal value of the function and observe how close three iterations takes the GD algorithm to the optimal point.
3. Verify single iteration convergence using Newton's method.

Question 5: Let $C \subset \mathbb{R}^2$ be the set of points (x, y) satisfying

$$x^2 + 2y^2 \leq 1, \quad x + y \leq 1, \quad y \leq x.$$

Consider the optimization problem

$$\min_{(x,y) \in C} f(x, y) = x + 3y.$$

1. State the KKT conditions for this problem.
2. Solve the KKT system and find the minimizer (x^*, y^*) and the minimum value $f(x^*, y^*)$.
3. Verify that the multipliers satisfy complementary slackness and nonnegativity.

Question 6: Verify the convexity of the functions below using the test for convexity.

1. $f(x) = \|x\|_\infty = \max_i |x_i|$.
2. $f(x) = e^{2x} - 4x \quad (x \in \mathbb{R})$.
3. $f(x) = x \log x \quad \text{for } x > 0$.

Hint:

- (Convexity Definition): $f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$.
- (First-Order Condition): $f(y) \geq f(x) + \nabla f(x)^\top (y - x) \quad \forall x, y$.
- (Second-Order Condition): $\nabla^2 f(x) \succeq 0 \quad \forall x$.

Question 7: Use the simplex method to solve the following LP model:

$$\begin{aligned} \text{Maximize } & Z = 5x_1 + 4x_2 \\ \text{subject to } & 6x_1 + 4x_2 \leq 24, \\ & x_1 + 2x_2 \leq 6, \\ & -x_1 + x_2 \leq 1, \\ & x_2 \leq 2, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Question 8: A bakery produces two types of products: Cakes and Cookies. The requirements per unit of each product are:

Ingredient	Cakes (g/unit)	Cookies (g/unit)
Flour	500	300
Sugar	200	150
Butter	100	50

The net contribution (profit) per unit is, Cake- Rs.50 and Cookie- Rs.30.
Available inventory of ingredients, Flour- 20 kg, Sugar- 10 kg and Butter- 5 kg.

1. Formulate the problem as a Linear Programming Problem.
2. Sketch the feasible region.
3. Locate the point at which function attains maxima.
4. Determine how many cakes (x_1) and cookies (x_2) the bakery should produce to maximize profit.

Question 9: Consider the optimization problem,

$$\begin{aligned} \text{minimize } & f(x) = x^2 + 1 \\ \text{subject to } & (x - 2)(x - 4) \leq 0, \\ & x \in \mathbb{R}. \end{aligned}$$

1. Give the feasible set, the optimal value, and the optimal solution.
2. Sketch the objective function. On the same plot, show the feasible set, optimal point and value, and plot the Lagrangian versus x for a few positive values of λ .
3. Derive and sketch the Lagrange dual function g .
4. State the dual problem, and verify that it is concave maximization problem.
5. Find the dual optimal value and dual optimal solution.

Question 10: For the Optimization problem given below, submit the python code to compute the minimizer using the following methods.

$$\min_{x \in \mathbb{R}} f(x) = x^3 - 3x + 1.$$

- Gradient Descent Method:

$$x_{k+1} = x_k - \alpha f'(x_k).$$

- Newton's Method:

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}.$$

Use stopping criterion $|f'(x_k)| < \varepsilon$ with $\varepsilon = 10^{-6}$.

Find a suitable α for the Gradient Descent Algorithm.

1. Compare the number of iterations taken by each method.
2. Run both algorithms from each of the two initial points $x_0 = 2.0$ and $x_0 = -2.0$.
3. For each run, record:
 - The number of iterations until convergence,
 - The final iterate x^* reached and whether it is a local minimizer or maximizer (you may check $f''(x^*)$),
 - The final objective value $f(x^*)$.
 - Plot the iterates on the graph of $f(x)$.
 - Summarize and discuss:
 - Which method converged faster (fewer iterations) and why?
 - Did any method fail to converge or encounter $f''(x_k) = 0$? If so, explain the reason.
 - How sensitive is GD to the step size α ?

Code submission guidelines

1. Prepare your assignment exclusively in a Jupyter Notebook (.ipynb format).
2. Ensure all code cells are executed and their corresponding outputs are visible.
3. Name the notebook as following: $\{Rollnumber\}_{LA_MFDS_Assignment\ 4.ipynb}$